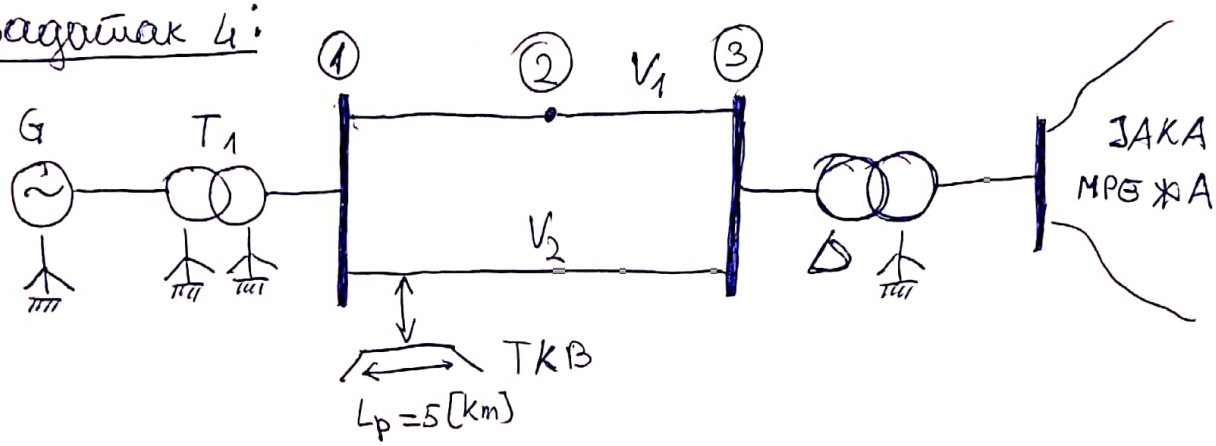
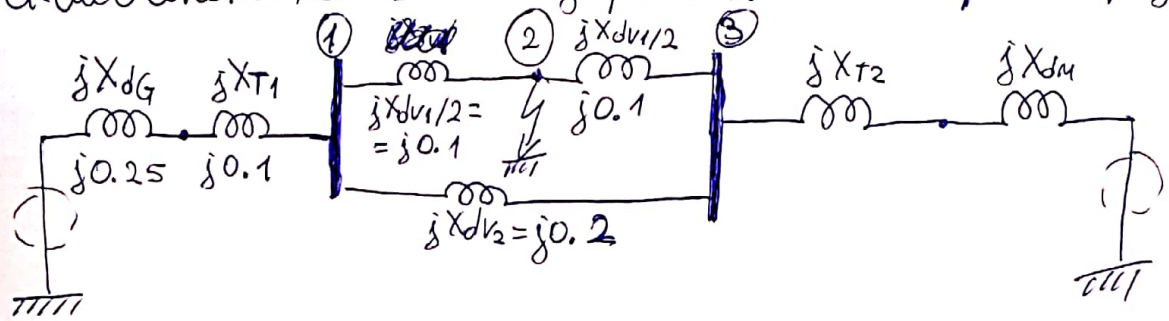
ЗАДАТАК ЗА ВЈЕЖБУ

22.04.2020.

Задатак 4:

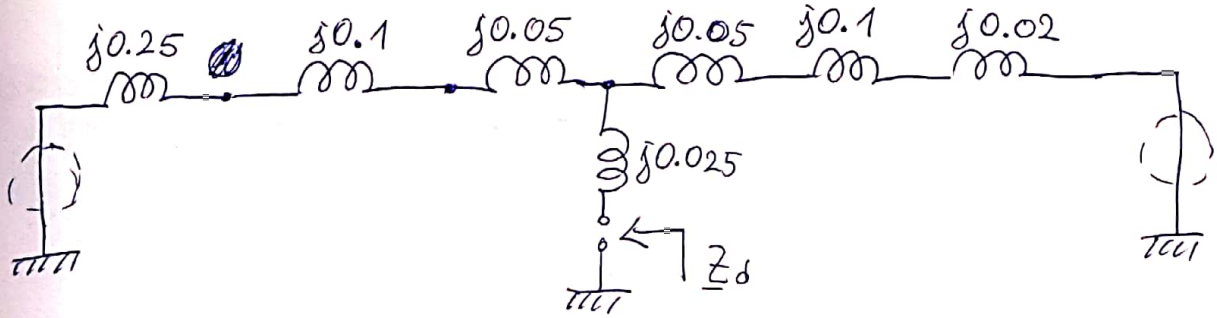


- Еквивалентна шема директног / инверзног редоследа



- Потребно је трансформисати троугао између сабирница 2 и сабирница 3 у звијезду

$$\underline{Z}' = \frac{j0.1 \cdot j0.2}{j0.1 + j0.1 + j0.2} = j0.05 \; [\Omega]; \quad \underline{Z}'' = \frac{j0.1 \cdot j0.1}{j0.1 + j0.1 + j0.2} = j0.025 \; [\Omega]$$



$$\underline{Z}_d = \underline{Z}_i = \frac{j(0.25 + 0.1 + 0.05) \cdot j(0.05 + 0.1 + 0.02)}{j(0.25 + 0.1 + 0.05 + 0.05 + 0.1 + 0.02)} + j0.025$$

$$\underline{Z}_d = \underline{Z}_i = j0.1443 \; [\Omega]$$

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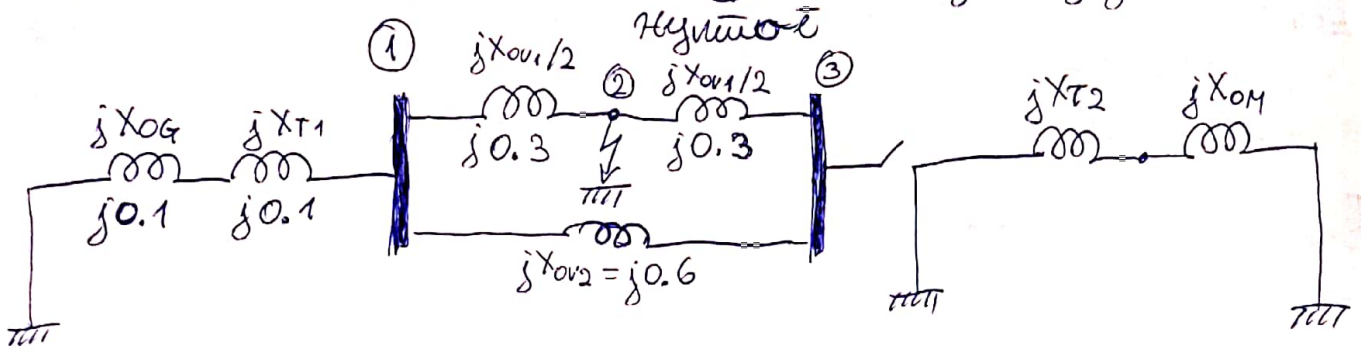
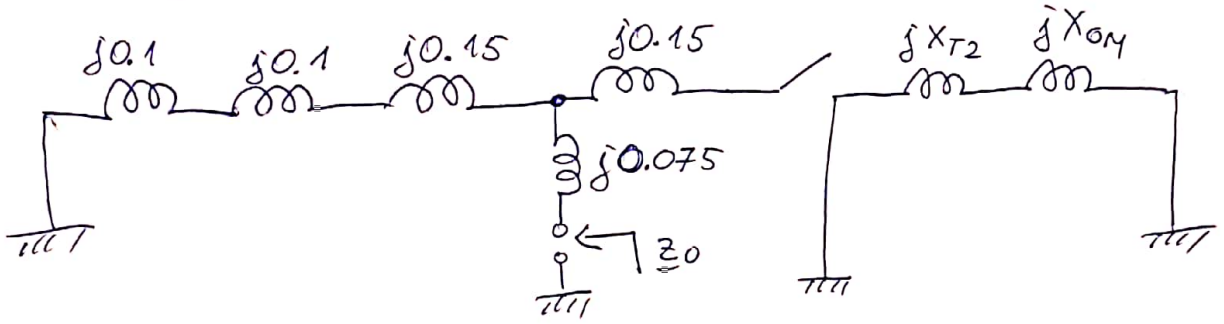
- Еквивалентна шема ~~[illegible]~~ редоследа нултог



• трансформација $\Delta \rightarrow Y$

$$\underline{Z}' = \frac{j0.3 \cdot j0.6}{j0.3 + j0.3 + j0.6} = j0.15\,[r.j.] \; ; \; \underline{Z}'' = \frac{j0.3 \cdot j0.3}{j0.3 + j0.3 + j0.6} = j0.075\,[r.j.]$$

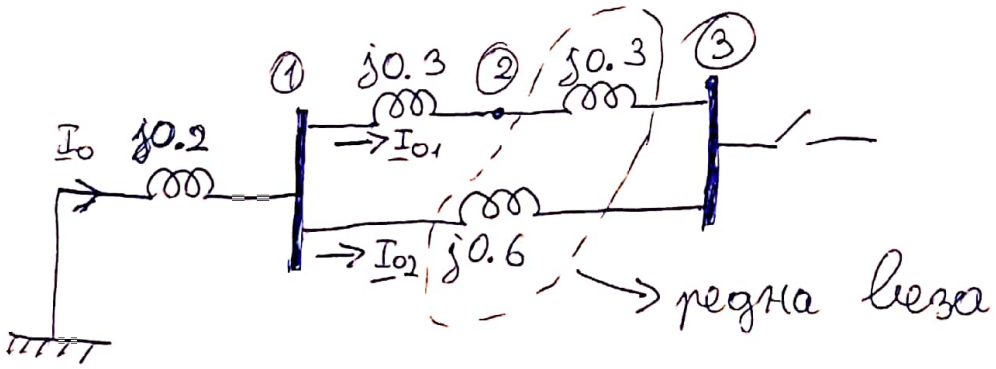


$$\underline{Z}_0 = j(0.1 + 0.1 + 0.15 + 0.075) = j0.4250\,[r.j.]$$

1PKS

$$\underline{I}_d = \underline{I}_i = \underline{I}_0 = \frac{\underline{U}_{fr}}{\underline{Z}_d + \underline{Z}_i + \underline{Z}_0} = \frac{1}{j(0.1443 + 0.1443 + 0.4250)}$$

$$\underline{I}_d = \underline{I}_i = \underline{I}_0 = -j1.4013\,[r.j.]$$



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$$\underline{I}_{ov1} = \frac{j(0.3+0.6)}{j(0.3+0.3+0.6)}\,\underline{I}_0 = \frac{j0.9}{j1.2}\cdot(-j1.4013) = -j1.0510\,[r.j.]$$

$$\underline{I}_{ov2} = \frac{j0.3}{j(0.3+0.3+0.6)}\,\underline{I}_0 = \frac{j0.3}{j1.2}\cdot(+j1.4013) = -j0.3503\,[r.j.]$$

$$I_B = \frac{S_B}{\sqrt{3}\,U_B} = \frac{100}{\sqrt{3}\cdot 220} = 0.2624\,[kA]$$

$$|\underline{I}_{ov1}|\,[kA] = |\underline{I}_{ov1}|\,[r.j.]\cdot 0.2624 = 1.0510\cdot 0.2624 = 0.2758\,[kA]$$

$$|\underline{I}_{ov2}|\,[kA] = |\underline{I}_{ov2}|\,[r.j.]\cdot 0.2624 = 0.3503\cdot 0.2624 = 0.0919\,[kA]$$

$$E_{v1} = 3\,|\underline{I}_{ov1}|\,[kA]\cdot 2\pi f\cdot L_p\cdot\frac{4}{\sqrt{a_1\,[m]}} = 3\cdot 0.2758\cdot 2\pi\cdot 50\cdot 5\cdot\frac{4}{\sqrt{450}}$$

$$E_{v1} = 245.07\,[V]$$

$$E_{v2} = 3\,|\underline{I}_{ov2}|\,[kA]\cdot 2\pi f\cdot L_p\cdot\frac{4}{\sqrt{a_2\,[m]}} = 3\cdot 0.0919\cdot 2\pi\cdot 50\cdot 5\cdot\frac{4}{\sqrt{400}}$$

$$E_{v2} = 86.61\,[V]$$

$$E = E_{v1} + E_{v2} = 245.07 + 86.61 \Rightarrow \boxed{E = 331.68\,[V]}$$

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